Ivan Sepulveda
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Education

University of San Francisco Bachelor of Science - Physics

Experience

Aqua Finance, Inc. - Remote *February 2026 - Present*

Manager of Analytics, Counterparty Risk

- Built counterparty risk frameworks, KPIs, early-warning indicators, and predictive models to monitor credit and fraud risk across dealers and applications
- Manage and mentor a Junior Analyst, and partner with Credit, Operations, Sales, Underwriting, and Data Engineering to translate business needs into scalable risk monitoring solutions
- Develop automated SQL/Python analytics pipelines and reporting, improving timeliness, consistency, and scalability of portfolio risk monitoring

Solar Mosaic LLC - Remote/Oakland, CA *June 2021 - February 2026*

Sr. Risk Analyst

- Developed SQL/Gephi network analysis using IP, geographic, and business-registration data to identify risk and fraud among connected entities
- Built an XGBoost dealer offboarding model to predict each dealer's probability of becoming offboarded at different time horizons (3M, 6M, 12M)
- Implemented dynamic partner risk scorecards, daily-updating schemas, alerts, and data initiatives with Data Engineering using `DBT, Apache Airflow, AWS, Snowflake`

L-Egant Solutions, LLC - Irvine, CA *October 2019 - June 2021*

Lead Software Developer [Contract]

- Developed scripts and GUI software for high-frame-rate camera operation, video capture, image algebra, pixel-intensity analysis, and two-dimensional Gaussian characterization

Mechanics Bank Auto Finance - Irvine, CA *August 2019 - June 2021*

Data Analyst - Portfolio & Servicing Risk

- Applied K-means, hierarchical clustering, and Random Forest models to segment collection agents, analyze credit metrics, inform credit policy, and forecast portfolio losses

Projects

when-where *2026*

Full-Stack Travel Recommendation Engine travel.iesepulveda.com

- **Data Sources:** REST APIs (World Bank Data360, Eurostat, Open-Meteo, Overpass) and web scraping (MICHELIN Guide) to build destination-scoring datasets
- **Pipeline:** Python ETL to clean, merge, and reconcile heterogeneous structured/unstructured data (weather, economic, crime, tourism) across inconsistent country/city coding schemes

dft (Duty Free Tracker) *2026*

Full-Stack Airport Shopping Data Platform dft.iesepulveda.com

- **Frontend:** Next.js/TypeScript — user login, favorites, interactive maps, image upload
- **Backend:** Supabase/Postgres — relational schema (Airport, Store, Sighting, Product, Brand)

Publications

[1] C. T. Chavez et al., "Measurement of 2D density profiles using a second-harmonic, dispersion interferometer," https://doi.org/10.1063/5.0119896

[2] M. Nikolic et al., "Applicability of optical emission spectroscopy techniques for characterization of Ar and Ar/O2 discharges," https://dx.doi.org/10.1088/1361-6463/abf61c

Skills

- **Languages & Libraries:** Python, SQL, R, LaTeX, Java, Matplotlib, Mayavi, ggplot2, Tableau, pandas, scikit-learn,
- **Tools & Platforms:** DBT, Airflow, Docker, AWS, Snowflake, GitHub, Bitbucket, Gephi
- **Methods:** Web Scraping, Network Analysis, Monte Carlo Simulations, XGBoost, Random Forest, KNN/K-Means